



WPYEG • Edmonton WordPress Meetup

Better by Default

Secure defaults for every WordPress site.

a hands-on workshop · build the “sane-defaults” plugin

WordPress is open by default; hosts vary in what they close.

None of these are bugs. They are defaults chosen for maximum compatibility on a 20+ year-old web application — you probably don't need them.



Username leak

REST + author archives list every login name to anonymous visitors.



Legacy XML-RPC wide open

Pingback and system.multicall amplifiers answer by default.



Dead weight loads

Emoji scripts, version tags, and RSD links on every page.



Spam surface invites

Comments, pingbacks, and trackbacks open by default.

The one idea to take home

A “default” is just an opinionated filter behind a toggle.

```
if ( wpyeg_defaults_enabled( 'restrict_rest_user_discovery' ) ) {  
    add_filter( 'rest_endpoints', $hide_users_endpoint );  
} // that's the whole pattern, repeated ~20 times
```

Hooks & filters

WordPress is built to be interrupted at labeled moments (hooks) so you never edit core code.



Action

“When you reach this moment, also DO this.”

```
add_action( 'init', fn );
```



Filter

“Before you use this value, let me CHANGE it first.”

```
add_filter( 'xmlrpc_enabled',  
    '__return_false' );
```

Six categories of defaults

1

Security

Shrink the attack surface

2

Content

Close spam channels & info leaks

3

Admin UX

A calmer, faster dashboard

4

Login

Sessions & credentials

5

Branding

Own the login screen

6

Performance

Trim the page weight

1

SECTION ONE

Security & Attack Surface

Every item here removes something an attacker can poke — usually in one line.

Restrict REST API user discovery

The `/wp/v2/users` endpoint hands out every author's login name to anyone — half of a brute-force guess, for free. Closing it for logged-out requests only keeps the editor and legit integrations working.

OPTION `wpyeg_restrict_rest_user_discovery`
DEFAULT
`yes`

```
add_filter( 'rest_endpoints', function ( $ep ) {  
    if ( ! is_user_logged_in() ) {  
        unset( $ep['/wp/v2/users'] );  
        unset(  
            $ep['/wp/v2/users/(?P<id>[\d]+)']  
        );  
    }  
    return $ep;  
} );
```

Lock REST to logged-in users (opt-in)

The sledgehammer version of the slide before: requiring auth for ALL REST calls stops anonymous scraping cold. It breaks anonymous REST — front-end blocks, embeds, search, outside integrations — so it ships off.

OPTION `wpyeg_disable_rest`
DEFAULT
`no`

```
add_filter( 'rest_authentication_errors',  
function ( $result ) {  
    if ( ! empty( $result ) ) return $result;  
    if ( ! is_user_logged_in() ) {  
        return new WP_Error(  
            'rest_not_logged_in', 'Auth required.',  
            array( 'status' => 401 ) );  
    }  
    return $result;  
} );
```


Lock XML-RPC down by category

XML-RPC isn't one door — it's an old switchboard, and every method is a phone line. Rather than rip the box off the wall, we unplug lines by category. Four switches, all off by default.

```
OPTION    wpyeg_xmlrpc_allow_* +  
_block_xmlrpc_endpoint  
DEFAULT  
no (all four)
```

```
// each category off → remove its methods  
add_filter( 'xmlrpc_methods', function ( $m ) {  
    if ( ! allow( 'pingbacks' ) )  
        unset( $m['pingback.ping'] );  
    if ( ! allow( 'remote_publishing' ) )  
        // drop wp.* metaWeblog.* mt.* blogger.*  
        return $m;  
} );  
  
// multicall can't be filtered off (IXR re-adds it)  
// → swap in a server that refuses it  
add_filter( 'wp_xmlrpc_server_class', $refuse );
```

Keep Application Passwords available

This is an existing default we don't lock down. An Application Password is like a spare key cut for one app: hashed, per-application, revocable on its own — the safer REST credential, and the only one core accepts for REST Basic Auth.

```
OPTION    wpyeg_disable_application_passwords
DEFAULT
no (available)
```

```
// available by default -
// prohibit only if opted in
if ( wpyeg_defaults_enabled(
    'disable_application_passwords' ) ) {
    add_filter(
        'wp_is_application_passwords_available',
        '__return_false'
    );
}
```

Require strong passwords

NIST 800-63B and OWASP now say length plus breach screening beats composition rules. So we require 15+ characters and screen every new password against Have I Been Pwned — all enforced server-side.

```
OPTION    wpyeg_require_strong_passwords
DEFAULT
yes
```

```
// hooked on user_profile_update_errors
if ( strlen( $pw ) < 15 ) {
    $errors->add( 'short', 'Use 15+ characters.' );
}

// screen against known breaches (HIBP) —
// length + screening, not composition rules
if ( wpyeg_password_is_pwned( $pw ) ) {
    $errors->add( 'pwned', 'Seen in a breach.' );
}
```

Remove fingerprints, add headers

Two headers with no real downside ship on; framing is its own setting, because it is the only one that can break a working site. Hiding the version is obscurity, not hardening, so it ships off.

```
OPTION    wpyeg_security_headers (yes) /  
wpyeg_frame_options (SAMEORIGIN)  
DEFAULT  
on / separate
```

```
remove_action( 'wp_head', 'wp_generator' );  
  
add_filter( 'wp_headers', function ( $h ) {  
    // only fill in what nothing else set  
    if ( ! isset( $h['X-Content-Type-Options'] ) )  
        $h['X-Content-Type-Options'] = 'nosniff';  
    if ( ! isset( $h['Referrer-Policy'] ) )  
        $h['Referrer-Policy'] =  
            'strict-origin-when-cross-origin';  
    return $h;  
} );  
// framing is its own setting - see the notes
```



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SECTION TWO

Content & Public Surfaces

Close the spam funnels and the thin pages Google (and bots) love to crawl.

Disable comments, trackbacks & pingbacks

For many sites, comments are a spam magnet with little upside. Here we close them everywhere, hide existing threads, and drop the admin menu.

```
OPTION    wpjeg_disable_comments
DEFAULT
yes
```

```
add_filter( 'comments_open', '__return_false', 20 );
add_filter( 'pings_open',    '__return_false', 20 );
add_filter( 'comments_array',
            '__return_empty_array', 20 );

// + remove_post_type_support() on init
// + remove_menu_page('edit-comments.php')
// + drop the admin-bar comments node
```

Redirect author & attachment pages

Author archives expose the authors' usernames in the URL, and attachment pages are near-empty media wrappers. Both dilute SEO and are targets for trouble. Same hook, two conditions.

```
OPTION    wpyeg_disable_author_archives /  
_redirect_attachment_pages  
DEFAULT  
yes / yes
```

```
add_action( 'template_redirect', function () {  
    if ( is_author() ) {  
        wp_safe_redirect( home_url('/'), 301 );  
        exit;  
    }  
    if ( is_attachment() ) {  
        // parent post, else the FILE - never home  
    }  
} );
```

Disable the emoji script

WordPress core injects an emoji-detection script and inline CSS on every page load, plus a DNS-prefetch hint. Modern browsers render emoji natively, so this is pure dead weight.

OPTION `wpyeg_disable_emojis`
DEFAULT
`yes`

```
add_action( 'init', function () {  
    remove_action( 'wp_head',  
        'print_emoji_detection_script', 7 );  
    remove_action( 'wp_print_styles',  
        'print_emoji_styles' );  
    // ...admin + feed + mail variants too  
    add_filter( 'emoji_svg_url',  
        '__return_false' );  
} );
```




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SECTION THREE

Admin UX & Login Sessions

Small quality-of-life defaults: a calmer dashboard and sensible session policy.

Faster search, quieter admin bar

Search the admin post list on a big site and WordPress reads every word of every post — like finding a book by reading the whole library. Title-only search checks just the spines, and it's far faster.

```
OPTION    wpyeg_title_only_admin_search /  
_frontend_admin_bar_behavior  
DEFAULT  
no / ''
```

```
// title-only admin search — narrow the COLUMNS  
add_filter( 'post_search_columns',  
    function ( $cols, $s, $q ) {  
        if ( is_admin() && $q->is_main_query() )  
            return array( 'post_title' );  
        return $cols;    // front-end untouched  
    }, 10, 3 );  
// hide bar for non-admins  
add_filter( 'show_admin_bar', fn( $s ) =>  
    current_user_can('manage_options') ? $s : false );
```

Right-size the login session

Click “Remember Me” and you stay logged in for 14 days. Cap that extended session, optionally shorten the regular one, or hide the checkbox entirely. One filter covers all three.

```
OPTION    wpyeg_remember_me_days /  
_session_regular_hours  
DEFAULT  
5 / 0
```

```
add_filter( 'auth_cookie_expiration',  
function ( $exp, $uid, $remember ) {  
    if ( $remember ) {  
        return 5 * DAY_IN_SECONDS;  
    }  
    return 12 * HOUR_IN_SECONDS;  
}, 10, 3 );  
  
// DAY_IN_SECONDS: core time constant
```



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SECTION FOUR

Branding & Performance

Own the login screen, then shave the last bit of weight off every page.

Own the login screen

The login page is a site's staff entrance, and by default the welcome mat links to someone else's house — that “W” points out to wordpress.org. Changing it uninvited is intrusive, so the default is to leave it alone.

```
OPTION    wpyeg_login_logo_behavior
DEFAULT
keep_default (keep / remove / unlink / replace)
```

```
// remove, unlink, or replace — a choice
add_action( 'login_head', $logo_css );

// any change points the link home
// (no separate toggle)
add_filter( 'login_headerurl', 'home_url' );
add_filter( 'login_headertext', fn() =>
    get_bloginfo( 'name' ) );
```

Throttle Heartbeat — and a default we deleted

Throttle Heartbeat to ease up on weak shared hosting. The more interesting half is the toggle that used to be here: WordPress 6.3 gave scripts a per-script loading strategy, so our blanket defer filter had to go.

```
OPTION    wpypg_throttle_heartbeat
DEFAULT
no (opt-in)
```

```
add_filter( 'heartbeat_settings', fn( $s ) => {
    $s['interval'] = 60; return $s;
} );

// Deferring is NOT a setting here. Since WP 6.3:
wp_enqueue_script( 'front', $src, array(), '1.0',
    array( 'strategy' => 'defer' ) );
```

Three things a plugin can't toggle

Some defaults live in wp-config.php, above the plugin layer, because they must load before plugins do. Document them as manual steps in your onboarding checklist.

```
define( 'DISALLOW_FILE_EDIT', true ); // no in-dashboard code editor
define( 'AUTOSAVE_INTERVAL', 120 ); // gentler autosave (seconds)
define( 'WP_POST_REVISIONS', 10 ); // cap revision-table bloat
```

1

Kills the theme/plugin editor

A stolen admin login can't rewrite your PHP.

2

Writes to the DB less often

Fewer autosave revisions during long edits.

3

Keeps revisions in check

Ten per post instead of unbounded growth.

How the plugin is built

One array is the whole map. Read it and you understand the plugin.

1 `schema()`

One array: every setting, its default, type & group. The single source of truth.



2 `settings page`

Loops the schema to render toggles under Settings → Better by Default.



3 `bootstrap()`

For each ENABLED key, wires its `add_filter` / `add_action` to the right hook.

```
$stored = get_option('wpyeg_better_by_default'); // read once
foreach ( wpyeg_defaults_schema() as $key => $field ) { /* render + wire */ }
```




Hands-on: install & flip switches

1

Upload the plugin

Plugins → Add New → Upload Plugin → choose sane-defaults.zip → Activate.

2

Open the settings

Settings → Better by Default; every toggle grouped by category.

3

Verify a default

Visit `/wp-json/wp/v2/users` logged out → 401 or empty, not a list of usernames.

4

Toggle & re-check

Flip a switch off, reload, watch the behavior change.

```
prefer the terminal? wp plugin install ./sane-defaults.zip --activate
```

Your turn

Add one new default to the plugin

Goal: disable the WordPress dashboard “Welcome” panel. Two small edits — no new settings-page code.

```
// 1) add a schema entry in wpyeg_defaults_schema()
'hide_welcome_panel' => array(
    'default' => 'yes', 'type' => 'toggle', 'group' => 'ux',
    'label' => 'Hide dashboard welcome panel',
),

// 2) wire it inside wpyeg_defaults_bootstrap()
if ( wpyeg_defaults_enabled( 'hide_welcome_panel' ) ) {
    remove_action( 'welcome_panel', 'wp_welcome_panel' );
}
```

The cheat sheet — defaults that ship ON

Everything on-by-default in one view, mapped to the core hook.

Default	Core hook	Category
Restrict REST user discovery	<code>rest_endpoints</code>	Security
Lock down XML-RPC by category	<code>xmlrpc_methods</code> / <code>wp_xmlrpc_server_class</code>	Security
Require strong passwords (15+ / breach-screened)	<code>user_profile_update_errors</code>	Security
Send baseline security headers	<code>wp_headers</code>	Security
Disable comments & pingbacks	<code>comments_open</code> / <code>pings_open</code>	Content
Redirect author + attachment pages	<code>template_redirect</code>	Content / SEO
Disable emoji script	<code>init</code> (remove_action)	Performance



Thanks, WPYEG!

Set your defaults wisely.

Files: `sane-defaults.zip` · `wordpress-default-settings.md`

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